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SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1715

COURSE OUTCOME COVERED

***CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour

QUESTIONS: 5

Total Marks:30

Annexure A

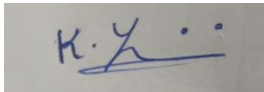
Resolution Algorithm Implementation

Code of Conduct:

I _K.Yamini(192572181)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

RUBRICS FOR CALCULATION

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Understanding of Problem Context	20% (2)	/2	/2	/2	/2	/2	/10
Application of Concepts	30% (3)	/3	/3	/3	/3	/3	/15
Problem-Solving Approach	20% (2)	/2	/2	/2	/2	/2	/10
Accuracy of Solution	20% (2)	/2	/2	/2	/2	/2	/10
Clarity	10% (1)	/1	/1	/1	/1	/1	/5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/50

1.RAIN AND WET GROUND

Given Knowledge Base:

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal:

Prove that **the ground is wet** using the Resolution Algorithm.

Algorithm:

Input: Knowledge Base and goal W.

Steps:

1. Define propositions R and W.
2. Represent the rule using propositional logic.
3. Convert the implication into CNF.
4. Add the given fact.
5. Negate the goal.
6. Apply resolution on clauses containing complementary literals.
7. Continue until the Empty Clause $\square$ is obtained.
8. If $\square$ is obtained, the goal is proved.

Step 1: Propositional Logic Representation

Let:

R = It is raining.

W = The ground is wet.

Rule:

$R \rightarrow W$

Fact:

R

Goal:

W

Step 2: CNF Conversion

Given:

$$R \rightarrow W$$

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Therefore:

$$\neg R \vee W$$

CNF Clauses

Clause	Expression
C1	$\neg R \vee W$
C2	R

Step 3: Negate the Goal

Goal:

$$W$$

Negated goal:

$$\neg W$$

Complete Clause Set

Clause	Expression
C1	$\neg R \vee W$
C2	R
C3	$\neg W$

Step 4: Resolution

Resolution 1

Resolve the selected clauses:

Clause 1: $\neg R \vee W$

Clause 2: R

Complementary literals: $\neg R$, R

Result: W

Therefore: $C4 = W$

Resolution 2

Resolve the selected clauses:

Clause 1: W

Clause 2: $\neg W$

Complementary literals: $W, \neg W$

Result: $\square$

Therefore: Empty Clause $\square$

Resolution Table

Step	Clause 1	Clause 2	Complementary Literals	Result
1	$\neg R \vee W$	R	$\neg R, R$	W
2	W	$\neg W$	$W, \neg W$	$\square$

Resolution Tree

$\neg R \vee W$ R
 \
 /
 \
 /
 W
 |
 $\neg W$
 |
 $\square$

Explanation

The knowledge base tells us that if it rains, the ground becomes wet. It is also given that it is raining. Using the first two clauses, resolution gives W . The goal was negated as $\neg W$. Resolving W and $\neg W$ produces the Empty Clause $\square$. This means that the negation of the goal leads to a contradiction.

Conclusion

Therefore, the ground is wet.

2 .STUDENT ASSIGNMENT SUBMISSION

Given Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks using the Resolution Algorithm.

Algorithm

Input: Knowledge Base and goal M.

1. Define the propositions S and M.
2. Represent the rule using propositional logic.
3. Convert the implication into CNF.
4. Add Rahul's submission as a fact.
5. Negate the goal.
6. Add the negated goal to the clause set.
7. Resolve the clauses containing complementary literals.
8. Derive the Empty Clause $\square$.
9. Conclude that Rahul receives internal marks.

Step 1: Propositional Logic Representation

Let:

S = Rahul submitted the assignment.

M = Rahul receives internal marks.

Rule:

$S \rightarrow M$

Fact:

S

Goal:

M

Step 2: CNF Conversion

Given:

$$S \rightarrow M$$

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Therefore:

$$\neg S \vee M$$

CNF Clauses

Clause	Expression
C1	$\neg S \vee M$
C2	S

Step 3: Negate the Goal

Goal:

$$M$$

Negated goal:

$$\neg M$$

Complete Clause Set

Clause	Expression
C1	$\neg S \vee M$
C2	S
C3	$\neg M$

Step 4: Resolution

Resolution 1

Resolve the selected clauses:

Clause 1: $\neg S \vee M$

Clause 2: S

Complementary literals: $\neg S, S$

Result: M

Therefore: $C4 = M$

Resolution 2

Resolve the selected clauses:

Clause 1: M

Clause 2: $\neg M$

Complementary literals: $M, \neg M$

Result: $\square$

Therefore: Empty Clause $\square$

Resolution Table

Step	Clause 1	Clause 2	Complementary Literals	Result
1	$\neg S \vee M$	S	$\neg S, S$	M
2	M	$\neg M$	$M, \neg M$	$\square$

Resolution Tree

$\neg S \vee M \quad S$
 \
 /
 \
 /
 M
 |
 $\neg M$
 |
 $\square$

Explanation

The knowledge base states that submitting the assignment results in receiving internal marks. Rahul submitted the assignment. Resolution derives M , while the negated goal gives $\neg M$. Resolving M and $\neg M$ produces the Empty Clause $\square$, proving the goal.

Conclusion

Therefore, Rahul receives internal marks.

3. LIBRARY MEMBERSHIP

Given Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

Prove that Priya can borrow books using the Resolution Algorithm.

Algorithm

Input: Knowledge Base and goal B.

1. Define the propositions L and B.
2. Represent the library rule using propositional logic.
3. Convert the implication into CNF.
4. Add the fact that Priya is a library member.
5. Negate the goal.
6. Add the negated goal to the clause set.
7. Resolve the resulting clauses.
8. Obtain the Empty Clause $\square$.
9. Conclude that Priya can borrow books.

Step 1: Propositional Logic Representation

Let:

L = Priya is a library member.

B = Priya can borrow books.

Rule:

$L \rightarrow B$

Fact:

L

Goal:

B

Step 2: CNF Conversion

Given:

$$L \rightarrow B$$

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Therefore:

$$\neg L \vee B$$

CNF Clauses

Clause	Expression
C1	$\neg L \vee B$
C2	L

Step 3: Negate the Goal

Goal:

B

Negated goal:

$\neg B$

Complete Clause Set

Clause	Expression
C1	$\neg L \vee B$
C2	L
C3	$\neg B$

Step 4: Resolution

Resolution 1

Resolve the selected clauses:

Clause 1: $\neg L \vee B$

Clause 2: L

Complementary literals: $\neg L$, L

Result: B

Therefore: C4 = B

Resolution 2

Resolve the selected clauses:

Clause 1: B

Clause 2: $\neg B$

Complementary literals: B, $\neg B$

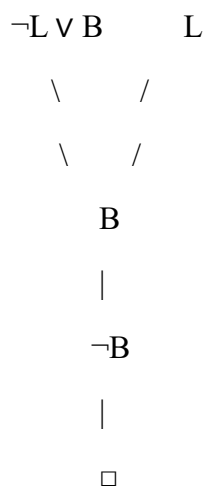
Result: $\square$

Therefore: Empty Clause $\square$

Resolution Table

Step	Clause 1	Clause 2	Complementary Literals	Result
1	$\neg L \vee B$	L	$\neg L, L$	B
2	B	$\neg B$	B, $\neg B$	$\square$

Resolution Tree



Explanation

The knowledge base states that a library member can borrow books. Priya is a library member. Resolution derives B, while the negated goal gives $\neg B$. Resolving B and $\neg B$ produces the Empty Clause $\square$, proving the goal.

Conclusion

Therefore, Priya can borrow books.

4. PLACEMENT ELIGIBILITY

Given Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

Prove that Arun is eligible for placement using the Resolution Algorithm.

Algorithm

Input: Knowledge Base and goal E.

1. Define the propositions A and E.
2. Represent the placement rule using propositional logic.
3. Convert the implication into CNF.
4. Add the fact that Arun cleared the aptitude test.
5. Negate the goal.
6. Add the negated goal to the clause set.
7. Resolve the clauses containing complementary literals.
8. Obtain the Empty Clause $\square$.
9. Conclude that Arun is eligible for placement.

Step 1: Propositional Logic Representation

Let:

A = Arun cleared the aptitude test.

E = Arun is eligible for placement.

Rule:

$A \rightarrow E$

Fact:

A

Goal:

E

Step 2: CNF Conversion

Given:

$A \rightarrow E$

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Therefore:

$$\neg A \vee E$$

CNF Clauses

Clause	Expression
C1	$\neg A \vee E$
C2	A

Step 3: Negate the Goal

Goal:

E

Negated goal:

$$\neg E$$

Complete Clause Set

Clause	Expression
C1	$\neg A \vee E$
C2	A
C3	$\neg E$

Step 4: Resolution

Resolution 1

Resolve the selected clauses:

Clause 1: $\neg A \vee E$

Clause 2: A

Complementary literals: $\neg A, A$

Result: E

Therefore: C4 = E

Resolution 2

Resolve the selected clauses:

Clause 1: E

Clause 2: $\neg E$

Complementary literals: $E, \neg E$

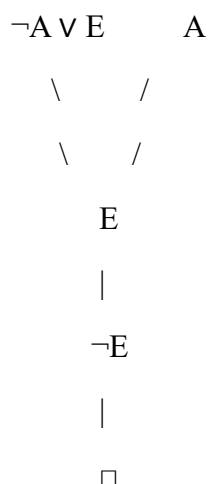
Result: $\square$

Therefore: Empty Clause $\square$

Resolution Table

Step	Clause 1	Clause 2	Complementary Literals	Result
1	$\neg A \vee E$	A	$\neg A, A$	E
2	E	$\neg E$	$E, \neg E$	$\square$

Resolution Tree



Explanation

The knowledge base states that a student who clears the aptitude test is eligible for placement. Arun cleared the aptitude test. Resolution derives E , while the negated goal gives $\neg E$. Resolving E and $\neg E$ produces the Empty Clause $\square$, proving the goal.

Conclusion

Therefore, Arun is eligible for placement.

5. ACCESS CONTROL SYSTEM

Given Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that the user is granted access using the Resolution Algorithm.

Algorithm

Input: Knowledge Base and goal G.

1. Define propositions P, A, and G.
2. Represent both rules using propositional logic.
3. Convert both implications into CNF.
4. Add the fact that the user entered the correct password.
5. Negate the goal.
6. Resolve the password rule with the password fact.
7. Obtain authentication.
8. Resolve the authentication rule with the authentication result.
9. Obtain granted access.
10. Resolve granted access with the negated goal.
11. Obtain the Empty Clause $\square$.
12. Conclude that the user is granted access.

Step 1: Propositional Logic Representation

Let:

P = User entered the correct password.

A = User is authenticated.

G = User is granted access.

Rule 1:

$P \rightarrow A$

Rule 2:

$A \rightarrow G$

Fact:

P

Goal:

G

Step 2: CNF Conversion

Rule 1:

$P \rightarrow A$

Therefore:

$\neg P \vee A$

Rule 2:

$A \rightarrow G$

Therefore:

$\neg A \vee G$

CNF Clauses

Clause	Expression
C1	$\neg P \vee A$
C2	$\neg A \vee G$
C3	P

Step 3: Negate the Goal

Goal:

G

Negated goal:

$\neg G$

Complete Clause Set

Clause	Expression
C1	$\neg P \vee A$
C2	$\neg A \vee G$
C3	P
C4	$\neg G$

Step 4: Resolution

Resolution 1

Resolve the selected clauses:

Clause 1: $\neg P \vee A$

Clause 2: P

Complementary literals: $\neg P, P$

Result: A

Therefore: $C4 = A$

Resolution 2

Resolve the selected clauses:

Clause 1: $\neg A \vee G$

Clause 2: A

Complementary literals: $\neg A, A$

Result: G

Therefore: $C5 = G$

Resolution 3

Resolve the selected clauses:

Clause 1: G

Clause 2: $\neg G$

Complementary literals: $G, \neg G$

Result: $\square$

Therefore: Empty Clause $\square$

Resolution Table

Step	Clause 1	Clause 2	Complementary Literals	Result
1	$\neg P \vee A$	P	$\neg P, P$	A
2	$\neg A \vee G$	A	$\neg A, A$	G
3	G	$\neg G$	$G, \neg G$	$\square$

Resolution Tree

$\neg P \vee A \quad P$

$$\begin{array}{c}
 \backslash \quad / \\
 \backslash \quad / \\
 A \\
 | \\
 \neg A \vee G \\
 | \\
 G \\
 | \\
 \neg G \\
 | \\
 \square
 \end{array}$$

Explanation

The user entered the correct password, so the first rule gives authentication. From authentication, the second rule gives granted access. The goal G is negated as $\neg G$. Resolution produces G and then resolves G with $\neg G$ to produce the Empty Clause $\square$. Therefore, the goal is proved.

Conclusion

Therefore, the user is granted access.